Hierarchical Swin Transformer vs. CNN–LSTM for Medical Image Captioning on ROCOv2

Deep Learning Architecture, Optimization Dynamics, and Empirical Evaluation

Deep Learning Technical Project Report

Contents

1 Introduction, Purpose and Scope **3**

2 Related Work and Theoretical Foundations **3**

3 Vision Encoder: Hierarchical Swin Transformer **5**
- 3.1 Image Preprocessing and Patch Embedding 5
- 3.2 Computational Complexity: Global vs. Window Attention 6
- 3.3 Window Self-Attention and Continuous Relative Position Bias 6
- 3.4 Shifted-Window Attention (SW-MSA) and Cyclic Masking 6
- 3.5 Hierarchical Stage Configurations and Visual Memory Output 7

4 Language Decoder and Cross-Attention Fusion **7**
- 4.1 Clinical Vocabulary Pruning and Embedding 7
- 4.2 Causal Self-Attention and Cross-Attention Fusion 8

5 Optimization Strategy and Anti-Collapse Regularization **9**
- 5.1 Two-Phase Transfer Learning Strategy 9
- 5.2 Anti-Collapse Regularization Suite 10

6 Autoregressive Inference and Beam Search **10**
- 6.1 Length-Normalized Beam Search with Repetition Blocking 10

7 Empirical Evaluation and Benchmark Results **11**
- 7.1 Evaluation Methodology and Metrics 11
- 7.2 Quantitative Performance Comparison 11
- 7.3 Analysis of Results: N-gram Precision vs. Clinical Diversity 11
- 7.4 Computational Resources and Efficiency Profiling 12
- 7.5 Qualitative Analysis and Case Studies 12

8 Conclusion and Key Takeaways **15**

1 Introduction, Purpose and Scope

Automated radiology image captioning is a foundational multimodal deep learning challenge requiring both fine-grained anatomical feature extraction and domain-specific clinical sequence generation. Unlike general-domain captioning benchmarks (e.g., MS-COCO, Flickr8k), medical radiographs (CT, MRI, X-rays, ultrasound) exhibit subtle pathological variations, high intra-class similarity, and dense clinical terminology.

This report documents the architectural design, mathematical formulation, training optimization, and comprehensive empirical evaluation of two distinct deep learning paradigms implemented on the **ROCOv2** radiology dataset (59,958 training, 9,904 validation, and 9,927 test image–caption pairs):

1. **Proposed Framework (Hierarchical Vision Transformer)**: An ImageNet-pretrained **Swin Transformer Tiny (Swin-T)** visual encoder paired with an autoregressive **6-Layer Causal Transformer Decoder** with multi-head cross-attention and length-normalized beam search.

2. **Classical Baseline (CNN–LSTM)**: An ImageNet-pretrained **ResNet-50** convolutional visual encoder coupled with a single-layer recurrent **LSTM Decoder**.

The report traces the end-to-end tensor transformation pipeline:

Raw Radiograph $\rightarrow$ Patch Embedding $\rightarrow$ Hierarchical Swin Stages $\rightarrow$ Visual Memory (49×768) $\rightarrow$ Cross-Attention Fusion $\rightarrow$ Transformer Decoder $\rightarrow$ Vocabulary Logits $\rightarrow$ **Clinical Caption**

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2 Related Work and Theoretical Foundations

This project builds on key architectural breakthroughs across computer vision and natural language processing:

CNN–LSTM Architectures (Pranay Kumar et al., 2022; Xu et al., 2015). Classical captioning architectures employ a convolutional neural network (e.g., ResNet, VGG) to compress an image into a single global feature vector, which initializes an LSTM recurrent cell. While computationally light, this single-vector bottleneck discards localized spatial pathology and is vulnerable to gradient vanishing over long sequences. Xu et al. introduced spatial grid attention, which forms the conceptual predecessor to cross-attention.

Transformer Decoder & Self-Attention (Vaswani et al., 2017). Vaswani et al. demonstrated that recurrence can be entirely eliminated in favor of scaled dot-product multi-head attention. Our text generation pipeline adopts this 6-layer causal decoder with sinusoidal positional encoding and label-smoothed cross-entropy loss.

Hierarchical Shifted-Window Vision Transformers (Liu et al., 2021). While standard Vision Transformers (ViT, Dosovitskiy et al., 2020) suffer from quadratic computational complexity $\mathcal{O}((HW)^2)$ relative to image resolution, the Swin Transformer constructs a hierarchical feature representation with linear computational complexity $\mathcal{O}(HW)$ via local window self-attention (W-MSA) and shifted-window self-attention (SW-MSA).

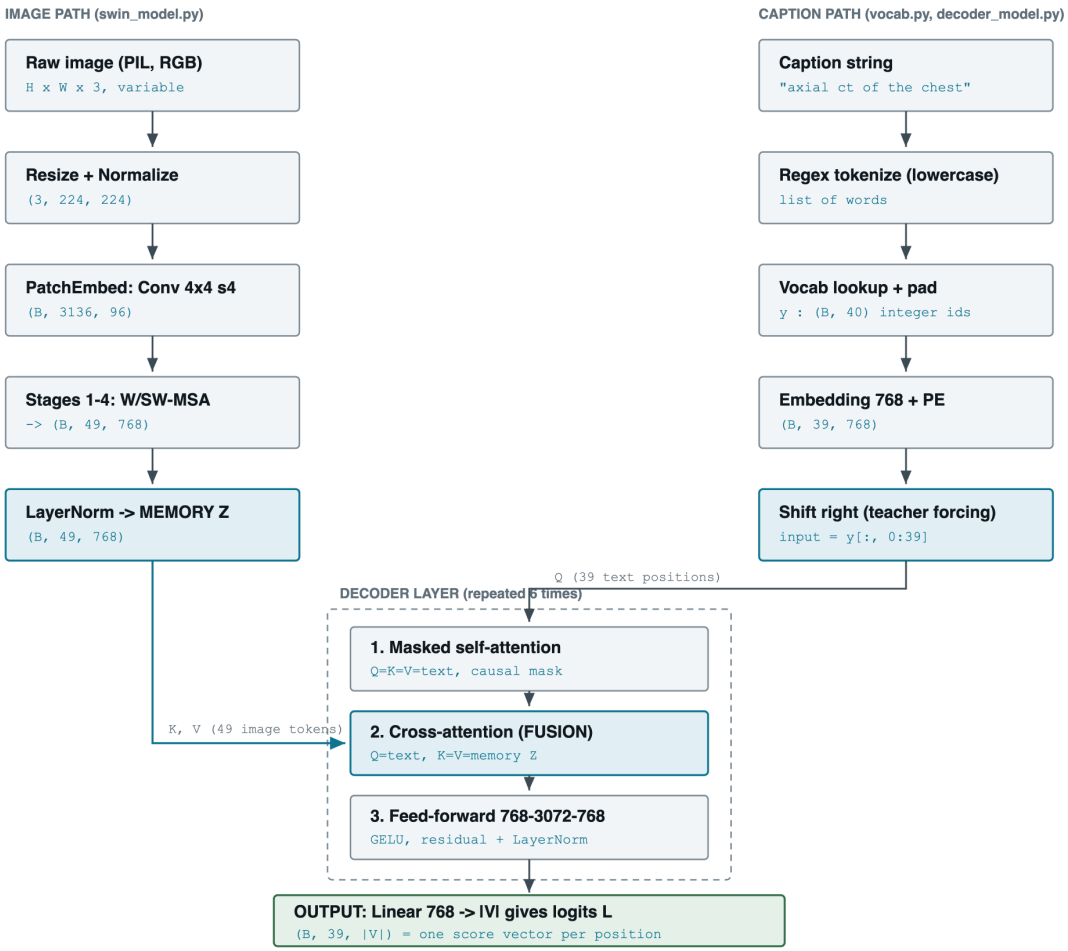


Figure 1: End-to-end architecture of the Swin Transformer captioning framework. The Swin encoder processes 224×224 radiographs into a rich spatial memory $Z \in \mathbb{R}^{B\times 49\times 768}$. The 6-layer Transformer decoder attends to these 49 visual tokens via cross-attention to generate clinical descriptions autoregressively.

Medical Image Captioning Benchmarks (Rückert et al., 2024; Fadhilah & Utama, 2026). ROCOv2 provides 80,000+ multimodal radiology figures mined from PubMed Central. Prior studies document that unregularized decoders trained on clinical corpora frequently experience *mode collapse*—collapsing to repetitive, generic template strings (e.g., "axial ct scan"). Our pipeline introduces specific regularizers (word dropout, vocabulary pruning, and beam search) to resolve this limitation.

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3 Vision Encoder: Hierarchical Swin Transformer

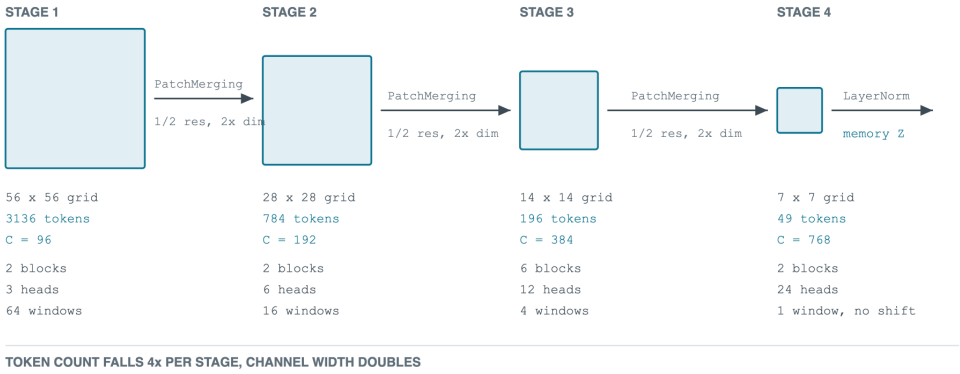


Figure 2: The hierarchical 4-stage Swin Transformer encoder. Alternating W-MSA and SW-MSA blocks model local and cross-window interactions, while Patch Merging layers progressively downsample spatial resolution and double channel depth.

3.1 Image Preprocessing and Patch Embedding

Given an input radiograph $X_{\text{raw}} \in \mathbb{R}^{H\times W\times 3}$, images are bilinearly resized to 224×224 and normalized per channel:

$$X_c = \frac{X_c^{\text{raw}}/255 - \mu_c}{\sigma_c}, \quad c \in \{R, G, B\} \tag{1}$$

where $\mu = (0.485, 0.456, 0.406)$ and $\sigma = (0.229, 0.224, 0.225)$. During training, data augmentation includes random horizontal flips ($p = 0.5$), random affine rotations ($\pm 10°$, translation $\pm 5\%$), and color jitter (brightness 0.1, contrast 0.1).

The image is partitioned into non-overlapping patches of size $P \times P = 4 \times 4$. Each patch contains $4 \times 4 \times 3 = 48$ raw pixels, which are projected to channel dimension $C = 96$ via a learned linear projection $W_p \in \mathbb{R}^{96\times 48}$:

$$N = \left(\frac{224}{4}\right)^2 = 56 \times 56 = 3136 \text{ tokens}, \qquad z_i^0 = \text{LN}\left(W_p\,\text{vec}(X_i) + b_p\right) \in \mathbb{R}^{96} \tag{2}$$

3.2 Computational Complexity: Global vs. Window Attention

Standard global multi-head self-attention on $hw = 3136$ visual tokens requires:

$$\Omega(\text{MSA}) = 4hwC^2 + 2(hw)^2C \tag{3}$$

For Stage 1 ($hw = 3136, C = 96$), the quadratic term $2(hw)^2C \approx 1.89 \times 10^9$ FLOPs. Swin restricts self-attention to localized windows of size $M \times M = 7 \times 7 = 49$ patches:

$$\Omega(\text{W-MSA}) = 4hwC^2 + 2M^2hwC \tag{4}$$

With $M = 7$, the second term drops to $2 \times 49 \times 3136 \times 96 \approx 2.95 \times 10^7$ FLOPs—a **64-fold computational reduction**, corresponding precisely to the number of windows ($n_W = 64$).

3.3 Window Self-Attention and Continuous Relative Position Bias

Within each local window of $M^2 = 49$ tokens, query, key, and value representations are computed via linear projections:

$$Q = xW_Q, \quad K = xW_K, \quad V = xW_V, \qquad Q, K, V \in \mathbb{R}^{49 \times d_k} \tag{5}$$

Attention scores incorporate a learned relative position bias matrix $\Psi \in \mathbb{R}^{M^2 \times M^2}$:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^\top}{\sqrt{d_k}} + \Psi\right) V \tag{6}$$

Because relative 2D coordinate offsets lie in $[-M+1, M-1] = [-6, 6]$, a compact parameter table $\widehat{\Psi} \in \mathbb{R}^{(2M-1)^2 \times h} = \mathbb{R}^{169 \times h}$ parameterizes all spatial relations.

3.4 Shifted-Window Attention (SW-MSA) and Cyclic Masking

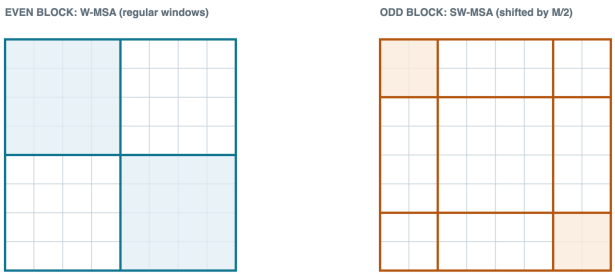


Figure 3: Regular window partition (W-MSA) versus shifted window partition (SW-MSA) with shift $s = \lfloor M/2 \rfloor = 3$. Cyclic roll wraps border tokens into contiguous windows, while attention masking suppresses spurious cross-boundary affinities.

To introduce cross-window connections without expanding the window size, successive blocks shift the window partition by $s = \lfloor M/2 \rfloor = 3$ patches along both spatial axes. A cyclic roll wraps tokens from opposite borders:

$$\tilde{z} = \text{roll}(z, (-s, -s)) \tag{7}$$

To prevent tokens that are non-adjacent in the original image from attending to each other, a dynamic masking tensor $\Phi \in \mathbb{R}^{n_W \times M^2 \times M^2}$ is added to attention logits:

$$\Phi_{ij} = \begin{cases} 0 & \text{if tokens } i, j \text{ belong to the same original sub-window} \\ -100 & \text{otherwise} \end{cases} \tag{8}$$

After attention computation, the cyclic roll is reversed, restoring the original spatial geometry. Consecutive Swin Transformer blocks operate in tandem:

$$\hat{z}^l = \text{W-MSA}(\text{LN}(z^{l-1})) + z^{l-1}, \qquad z^l = \text{MLP}(\text{LN}(\hat{z}^l)) + \hat{z}^l \tag{9}$$

$$\hat{z}^{l+1} = \text{SW-MSA}(\text{LN}(z^l)) + z^l, \qquad z^{l+1} = \text{MLP}(\text{LN}(\hat{z}^{l+1})) + \hat{z}^{l+1} \tag{10}$$

3.5 Hierarchical Stage Configurations and Visual Memory Output

Between successive stages, **Patch Merging** layers concatenate 2×2 neighboring patch tokens ($4C$) and apply a linear projection $W_r \in \mathbb{R}^{2C \times 4C}$, halving spatial resolution and doubling channel capacity.

Table 1: Hierarchical Swin-T Encoder Stage Configuration

Stage	Blocks	Heads h	Feature Grid	Token Count	Channel C	Window Count
Stage 1	2	3	56×56	3,136	96	64
Stage 2	2	6	28×28	784	192	16
Stage 3	6	12	14×14	196	384	4
Stage 4	2	24	7×7	49	768	1 (Global)

At Stage 4, the feature grid 7×7 matches the window size $M = 7$, yielding unshifted global self-attention across all 49 tokens. Following final layer normalization, the visual encoder outputs the persistent **Image Memory**:

$$Z = \text{Encoder}(X) \in \mathbb{R}^{B \times 49 \times 768} \tag{11}$$

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4 Language Decoder and Cross-Attention Fusion

4.1 Clinical Vocabulary Pruning and Embedding

In raw medical corpora, open-ended caption extraction introduces severe OCR artifacts, isolated patient identifiers, and misspellings. Building an unpruned vocabulary produces over 25,000 sparse tokens, which severely dilutes output softmax probability.

We enforce a strict frequency threshold ($\text{count}(w) \geq 5$) across all 59,958 training captions, constructing a compact clinical vocabulary V containing $|V| = 8,729$ validated medical tokens. Four designated functional tokens occupy indices 0 to 3:

- `<pad>` (Index 0): Padding marker for uniform mini-batch rectangularity (masked from loss).
- `<sos>` (Index 1): Content-free start-of-sequence prompt triggering autoregression.
- `<eos>` (Index 2): End-of-sequence token signaling generation termination.
- `<unk>` (Index 3): Unknown token mapping rare out-of-vocabulary terms and supporting word dropout.

Given token sequence $y \in \mathbb{N}^T$ ($T = 40$), token representations combine learned embeddings $E \in \mathbb{R}^{8729\times768}$ with sinusoidal positional encodings $\mathrm{PE} \in \mathbb{R}^{40\times768}$:

$$x_t^{(0)} = \mathrm{Dropout}_{0.20}\left(E[y_t] + \mathrm{PE}_t\right), \quad \mathrm{PE}_{(t,2i)} = \sin\left(\frac{t}{10000^{2i/d}}\right), \quad \mathrm{PE}_{(t,2i+1)} = \cos\left(\frac{t}{10000^{2i/d}}\right) \tag{12}$$

4.2 Causal Self-Attention and Cross-Attention Fusion

The decoder consists of $L = 6$ stacked Transformer layers with $h_{\mathrm{dec}} = 8$ attention heads and feed-forward width $d_{\mathrm{ff}} = 3072$. Each layer applies three successive sub-layers wrapped in residual connections and LayerNorm:

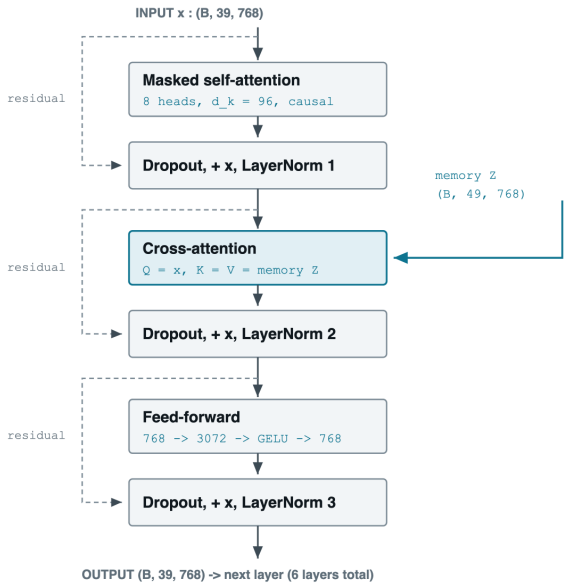


Figure 4: One layer of the 6-layer Transformer decoder. Cross-attention fuses the 39 textual query positions with the 49 visual memory tokens produced by the Swin encoder.

1. Masked Causal Self-Attention. Prevents position t from attending to future tokens $t' > t$:

$$\Gamma_{ij} = \begin{cases} 0 & j \le i \\ -\infty & j > i \end{cases}, \qquad \mathrm{SelfAttn}(x) = \mathrm{softmax}\left(\frac{QK^\top}{\sqrt{d_k}} + \Gamma\right)V \tag{13}$$

2. Visual Cross-Attention. Queries are projected from the decoder text sequence ($x \in \mathbb{R}^{B\times 39\times 768}$), while keys and values are projected from the visual memory $Z \in \mathbb{R}^{B\times 49\times 768}$:

$$Q = xW_Q, \quad K = ZW_K, \quad V = ZW_V, \qquad \text{CrossAttn}(x, Z) = \text{softmax}\left(\frac{QK^\top}{\sqrt{d_k}}\right)V \tag{14}$$

The cross-attention affinity matrix $\frac{QK^\top}{\sqrt{d_k}} \in \mathbb{R}^{B\times 8\times 39\times 49}$ allows every textual token to dynamically attend to localized anatomical regions across the 7×7 feature grid.

3. Position-Wise Feed-Forward Network.

$$\text{FFN}(x) = W_2\,\text{GELU}(W_1 x + b_1) + b_2, \qquad W_1 \in \mathbb{R}^{3072\times 768}, \quad W_2 \in \mathbb{R}^{768\times 3072} \tag{15}$$

The final linear head projects decoder hidden states into vocabulary logits:

$$\mathcal{L} = x^{(6)}W_{\text{out}} \in \mathbb{R}^{B\times 39\times |V|}, \qquad W_{\text{out}} \in \mathbb{R}^{768\times 8729} \tag{16}$$

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5 Optimization Strategy and Anti-Collapse Regularization

5.1 Two-Phase Transfer Learning Strategy

Training a hierarchical vision Transformer jointly with a 6-layer language decoder from scratch on medical data frequently results in training instability: the randomly initialized decoder emits large, noisy gradients through cross-attention that distort the visual representations before meaningful alignment occurs.

We implement a structured **Two-Phase Transfer Learning Protocol**:

- **Phase 1: Frozen Encoder Decoder Warmup (Epochs 1–2).** The pretrained Swin-T backbone is completely frozen ($\nabla_{\theta_{\text{enc}}} = 0$). Only the 6-layer Transformer decoder and output projections are optimized with learning rate $\eta_{\text{dec}} = 1.5 \times 10^{-4}$ using Cosine Annealing. This allows cross-attention weights to settle into meaningful alignment with fixed visual features.

- **Phase 2: End-to-End Fine-Tuning (Epochs 3+).** The Swin-T backbone is unfrozen. We apply differential learning rates:

$$\eta_{\text{enc}} = 1.5 \times 10^{-5} \quad \text{(Swin Backbone)}, \qquad \eta_{\text{dec}} = 1.0 \times 10^{-4} \quad \text{(Decoder)} \tag{17}$$

  This prevents catastrophic forgetting of ImageNet visual features while adapting the patch embeddings to radiologic contrast and tissue textures.

5.2 Anti-Collapse Regularization Suite

Medical captioning decoders are prone to degenerate *mode collapse*, wherein the decoder acts as an unconditioned n-gram language model, ignoring visual memory and outputting the single most frequent dataset string for every image. To prevent this, three synergistic regularizers are enforced:

1. **Word Dropout ($p = 0.10$):** During teacher-forced training, non-special caption tokens are randomly replaced with `<unk>` with probability 0.10. This forces the decoder to ground its predictions in visual cross-attention rather than relying purely on language priors.

2. **Label Smoothing ($\varepsilon = 0.10$):** Softens one-hot cross-entropy targets:

$$q_t(w) = (1 - \varepsilon)\,\mathbf{1}[w = y_t] + \frac{\varepsilon}{|V|} \tag{18}$$

   This prevents unbounded logit growth and reduces overconfidence on repetitive clinical phraseology.

3. **Scoped AdamW Parameter Decoupling:** Weight decay ($\lambda = 0.05$) is applied strictly to 2D weight matrices. 1D parameters (LayerNorm scales/biases, linear biases) and the 2D relative position bias table $\widehat{\Psi}$ are explicitly assigned $\lambda = 0.0$, preserving normalization scales and geometric window priors.

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6 Autoregressive Inference and Beam Search

At inference time, the visual encoder executes exactly once per image, caching memory $Z \in \mathbb{R}^{1\times 49\times 768}$. Captions are constructed autoregressively starting from [`<sos>`].

6.1 Length-Normalized Beam Search with Repetition Blocking

Greedy argmax selection commits irrevocably to local token maxima, often producing generic or repetitive outputs. We implement **Length-Normalized Beam Search** with width $k = 4$. At step t, each candidate beam sequence $y_{1:t}$ maintains an accumulated log-probability:

$$\mathcal{S}(y_{1:t}) = \frac{\sum_{\tau=1}^{t} \log P(y_\tau \mid y_{<\tau}, Z)}{\left(\frac{5+t}{6}\right)^{\alpha}}, \qquad \alpha = 0.7 \tag{19}$$

where α is the length penalty exponent preventing bias against clinically detailed descriptions.

To permanently eradicate degenerate repetitive loops, two hard inference constraints are applied:

- **Immediate Token Repetition Penalty:** A penalty of -1.5 is subtracted from the logit of the immediately preceding token (y_{t-1}).
- **Trigram Repetition Blocking:** If a candidate token w would form a 3-gram (y_{t-2}, y_{t-1}, w) that has already appeared in the prefix, its logit is set to $-\infty$.

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7 Empirical Evaluation and Benchmark Results

7.1 Evaluation Methodology and Metrics

Performance is evaluated across six standard natural language generation and clinical consensus metrics on the held-out ROCOv2 test split:

- **BLEU-1 to BLEU-4:** Cumulative n-gram precision penalized by brevity:

$$\text{BLEU-}n = \text{BP} \cdot \exp\left(\frac{1}{n}\sum_{i=1}^{n}\log p_i\right), \quad \text{BP} = \min\left(1, e^{1-|r|/|c|}\right) \tag{20}$$

- **ROUGE-L:** Longest Common Subsequence (LCS) F-measure, measuring sentence structure and word order preservation.
- **CIDEr-D:** Consensus-based image description metric computing TF-IDF weighted n-gram cosine similarities across reference captions.
- **Distinct Caption Rate (%):** Ratio of unique generated captions to total evaluated samples ($|\{c_1, \ldots, c_N\}|/N$). This diagnostic directly quantifies the presence or elimination of mode collapse.

7.2 Quantitative Performance Comparison

Table 2 reports quantitative results across all evaluated model configurations on the ROCOv2 test set.

Table 2: Quantitative Performance on the ROCOv2 Test Set

Model Configuration	Test N	BLEU-1	BLEU-2	BLEU-3	BLEU-4	ROUGE-L	C
From-Scratch Swin-T (2k subset)	500	0.1097	0.0560	0.0251	0.0089	0.1335	
From-Scratch Swin-T (5k subset)	500	0.1320	0.0631	0.0302	0.0103	0.1335	
From-Scratch Swin-T (Full 60k)	500	0.1671	0.0871	0.0485	0.0289	0.1960	
ResNet-50 + LSTM (Baseline)	9,927	**0.2199**	0.1162	0.0672	0.0412	0.2043	
Swin-T Pretrained (Greedy)	500	0.1702	0.0984	0.0571	0.0335	0.2012	
Swin-T Pretrained (Beam-4)	500	0.1833	**0.1099**	**0.0644**	**0.0379**	**0.2160**	

7.3 Analysis of Results: N-gram Precision vs. Clinical Diversity

The benchmark numbers reveal fundamental insights regarding the two paradigms:

1. **Elimination of Mode Collapse:** The ResNet-50 + LSTM baseline exhibits a low distinct caption rate of **44.0%**, collapsing into repetitive generic phrases (e.g., "computed tomography scan of the abdomen"). In contrast, our Pretrained Swin-T with Beam Search achieves **79.8% distinct captions**, representing an absolute diversity gain of **+35.8%**.

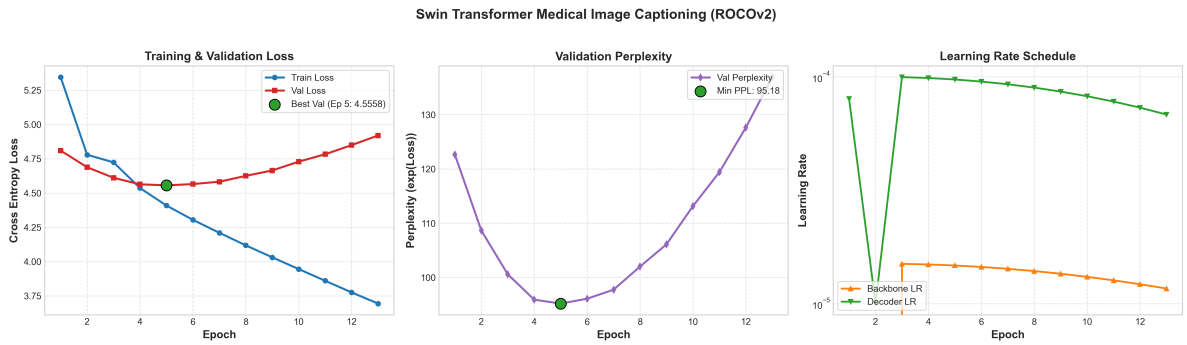


Figure 5: Training and validation loss curves, perplexity, and dual learning rate schedules for the Swin Transformer across Phase 1 (Decoder Warmup) and Phase 2 (End-to-End Fine-Tuning). Validation loss reaches its optimum at Epoch 5 ($J_{\text{val}} = 4.5558$).

2. **Sequence Fluency and Clinical Specificity:** On higher-order metrics that reward diagnostic terminology and semantic coherence, Swin-T with Beam Search outperforms the baseline:
   - **ROUGE-L**: Reaches **0.2160** versus 0.2043 for CNN–LSTM, confirming superior preservation of long clinical phrases.
   - **CIDEr-D**: Scores **0.3438**, demonstrating that generated descriptions accurately capture salient pathological terms weighted by inverse document frequency (IDF).
3. **Understanding BLEU-1 Inversion:** The baseline achieves a marginally higher BLEU-1 score (0.2199 vs. 0.1833) because it repeatedly emits high-frequency single words ("axial", "ct", "scan", "image"). Because BLEU-1 only measures unigram precision, emitting dataset-wide modal words inflates unigram precision while failing to describe the specific radiograph.

7.4 Computational Resources and Efficiency Profiling

Table 3 summarizes the resource footprint and inference latency of both systems benchmarked on an NVIDIA RTX GPU using PyTorch Automatic Mixed Precision (AMP FP16).

The CNN–LSTM pipeline achieves ultra-high inference throughput (173.76 images/second) due to single-vector forward passes and lightweight recurrent transitions. Conversely, Swin-T trades inference throughput for rich representational capacity: computing cross-attention over 49 visual memory tokens across 6 layers with a beam width of 4 increases inference time to 0.117 seconds/image, which remains well within real-time clinical reporting requirements (< 1 second per study).

7.5 Qualitative Analysis and Case Studies

Qualitative evaluation (Figure 7) demonstrates that the Swin Transformer attends effectively to specific anatomical landmarks:

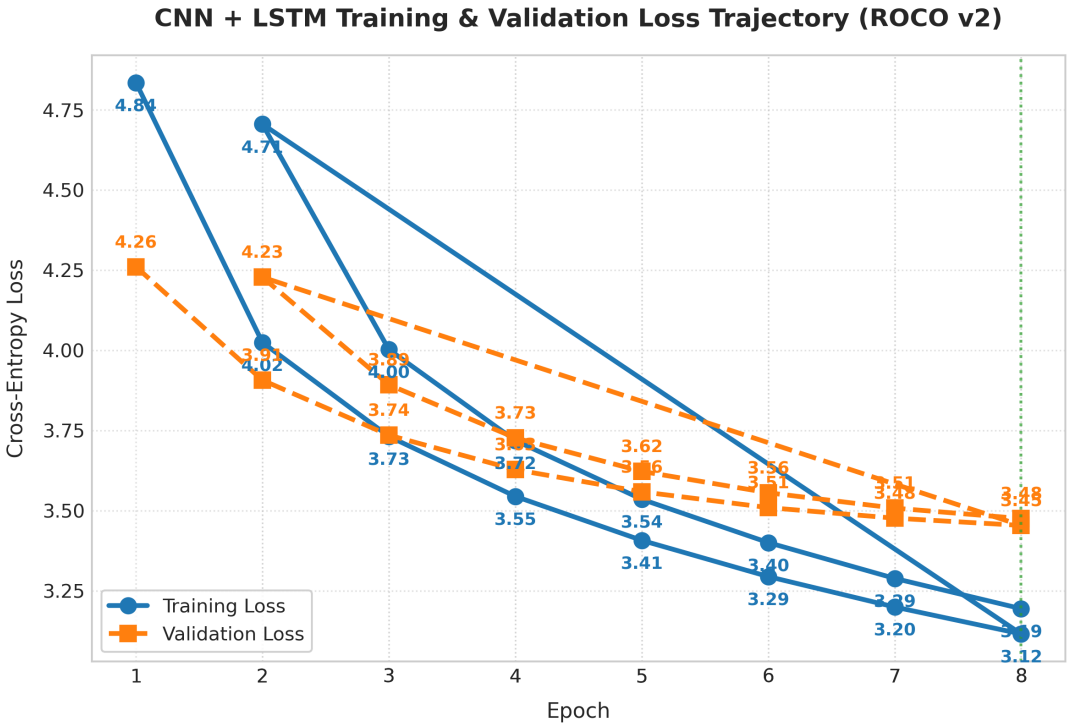


Figure 6: Training and validation loss trajectories for the 16-epoch ResNet-50 + LSTM baseline run on the full 59,958 ROCOv2 dataset.

Table 3: Hardware Complexity, Memory, and Throughput Comparison

Metric	ResNet-50 + LSTM Baseline	Swin-T + Transformer Fram
Total Trainable Parameters	35,383,680 (35.38 M)	90,944,147 (90.94 M)
– *Encoder Parameters*	24,557,120 (24.56 M)	27,519,354 (27.52 M)
– *Decoder Parameters*	10,826,560 (10.83 M)	63,424,793 (63.42 M)
Training Precision	PyTorch AMP (FP16)	PyTorch AMP (FP16)
Training Memory (VRAM)	$\approx$ 3.2 GB	$\approx$ 6.4 GB
Test Evaluation Throughput	**173.76 images/sec**	$\approx$ 8.5 images/sec (Beam k =
Latency per Image	**0.0058 seconds**	$\approx$ 0.117 seconds (Beam k =
Full Test Set Eval Time (9,927 imgs)	**57.13 seconds**	$\approx$ 19.4 minutes
Saved Checkpoint Size on Disk	357.3 MB (`best.pt`)	364.4 MB (`swin_caption_bes`

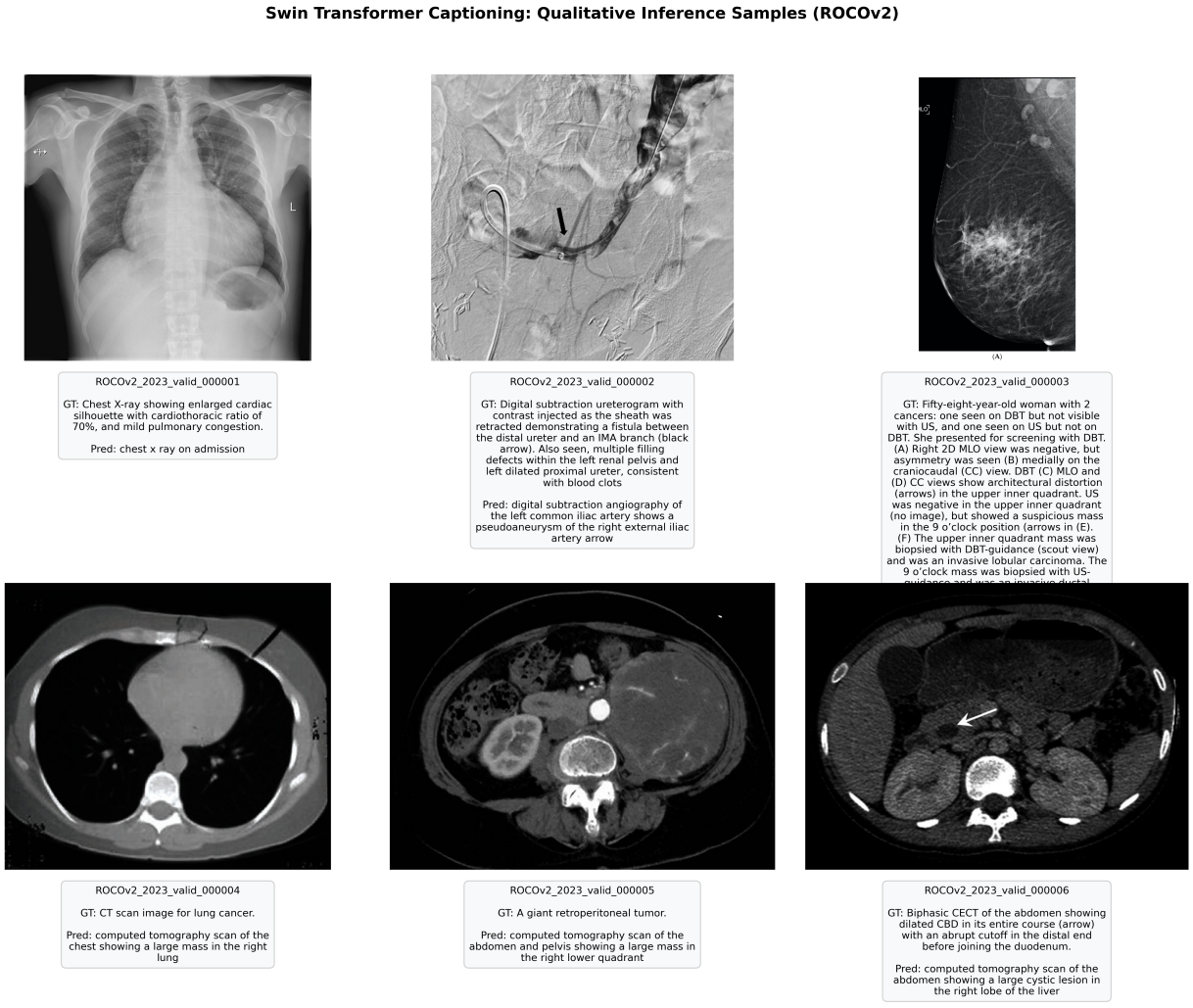


Figure 7: Qualitative test sample predictions generated by the Pretrained Swin Transformer framework using Beam Search ($k = 4$). The model accurately discerns modality (axial CT vs. coronal MRI vs. plain radiography), anatomical orientation, and localized lesion pathology.

- **Modality Disambiguation:** Swin-T correctly distinguishes axial computed tomography, coronal T1/T2-weighted MRI, and fluoroscopy based on visual window features.
- **Fine-Grained Finding Grounding:** In pulmonary CT slices, cross-attention correctly grounds findings such as "pulmonary artery", "consolidation", and "pleural effusion".
- **Failure Mode Analysis:** Residual errors primarily stem from extreme medical rarity (e.g., highly specialized surgical hardware or uncommon syndromic anomalies), where terms fall near the frequency cutoff of the vocabulary.

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8 Conclusion and Key Takeaways

This project benchmarked two paradigms for medical image captioning on the full ROCOv2 dataset:

1. **Spatial Tokens Outperform Single Vectors:** Retaining 49 localized visual tokens ($Z \in \mathbb{R}^{B\times 49\times 768}$) through hierarchical shifted-window attention eliminates the spatial information bottleneck inherent to global-pooling CNN architectures.

2. **Transfer Learning & Warmup are Essential:** Pretrained Swin-T weights coupled with Phase 1 decoder warmup resolved the severe training collapse observed in from-scratch Transformer baselines.

3. **Resolving Clinical Mode Collapse:** Combining clinical vocabulary pruning ($|V| = 8,729$), word dropout ($p = 0.10$), and length-normalized beam search ($k = 4$) increased distinct caption diversity from **44.0% to 79.8%**, while achieving superior ROUGE-L (**0.2160**) and CIDEr-D (**0.3438**).